

soncli.mcs: An After-Market Open-Source Enabler

An Evolutionary Bridge
From Nokia EdenNet SON to More Than SON

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Video File: ./video/mbIntro.mp4

Summary — Acts I and II

Act I: soncli.mcs as an After Market Story.

Vendor & Product

- Nokia EdenNet SON

Customers & Users

- T-Mobile, BT,
Orange, Telstra,
NTT DOCOMO, ...

Gap & Our Offerings

- soncli.mcs – A CLI
Interface for Running
SON Modules

Act II: Absorb EdenNet SON Into the Much Richer BISOS-PyCS Framework

Providing: Web-UI for soncli.mcs, Observability, Parallel Execution

Once Absorbed, the soncli.mcs derivative inherits:

- Web UI for soncli.mcs for GUI specification of parameters and targets
- Observability of execution and results
- Nation Wide Parallel Execution of Modules

Summary — Acts III and IV

Act III: Extending Targets Beyond RAN — With MaTOOF

- Through Having Been Absorbed into BISOS-PyCS, Needed Facilities for Realizing the Concept of Modules and Targets Oriented Open Fabric (MaTOOF) Universal are now in Place.
- Targets need not be limited to what Nokia supports.

Act IV: Make EdenNet SON Fully Your Own

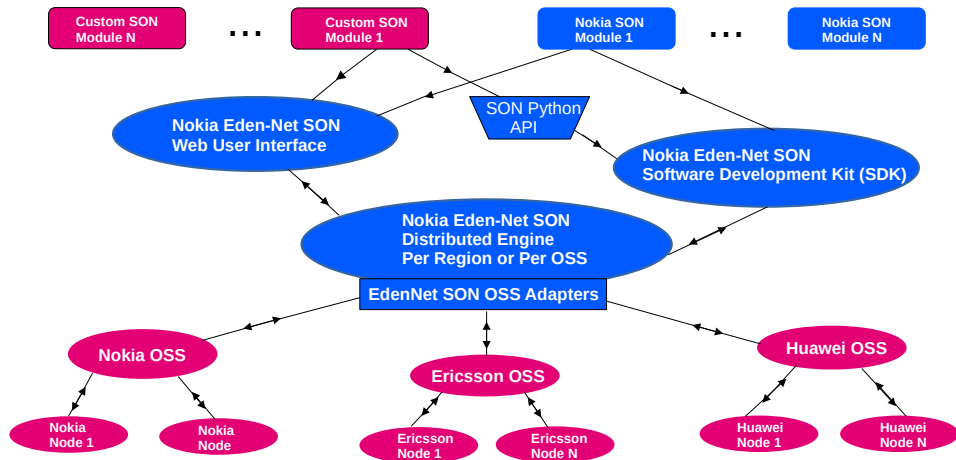
- Simply Replace OSS Interfaces with Your Own
- soncli.mcs and BISOS-PyCS already provide everything else that is needed.

Abstract Nokia EdenNet SON Parts List

Nokia EdenNet SON Parts List

- **SON Python Modules**
 - SON Python **Custom** Modules
 - SON Python **Nokia** Modules
- **SON Python API**
- **SON Python Software Development Kit (SDK)**
(The SDK is also an External Execution Facility — EEF)
- SON Web User Interface
- SON Execution Engine
- SON OSS Interface Adapters

Abstract Nokia EdenNet SON Ecosystem



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Publicly Known List of Nokia SON Clients

AI and Google Say: “Based on public announcements and reports, the following telecom companies use or have used Nokia EdenNet SON:”

- **T-Mobile USA**
- **Telstra** (Australia)
- **Orange** (Global/Europe/MEA)
- **Vodafone Idea** (India)
- **Telecom Argentina**
- **BT (EE)** (UK)
- **NTT DOCOMO** (Japan)

And probably many other telecom companies that have not been publicly disclosed.

Applicability of soncli.mcs to You

Intended Audience for This Presentation

Audience	Nokia SON Customers (T-Mobile, ...)	NOKIA	Telecom Operators (AT&T, ...)	Internet Providers (Google, ...)
SON Modules Developer:	X X X X X	X X X X	X X	— — —
SON Modules User:	X X X X X	X X X X	X X	— — —
O-RAN Management:	X X	X X	X X	X X X
Common Sys Management:	X X X	X X X	X X X	X X X

Strength and Weaknesses of the Nokia SON Architecture

Strength:

- Choice of Python (12 years ago)
- Modules Architecture for RAN Systems Management
- Targets Architecture for Cells
- Python API for RAN and Telecom Sys Mgmt Abstractions

Weaknesses:

- Jailed in the Telecom Walled Garden
- Abstraction of Targets is Limited to Cells
- Limited in Scope to RAN and Telecom
- Limited to a a single Modules Player (Web-UI)
- No CLI for Modules Execution

With soncli.mcs We Can Liberate Modules and Targets

- Inside of the Telecom Walled Garden Open Up and Extend EdenNet SON
- Profit from the Nokia SON Experience of “Modules and Target Oriented Open Fabric” (MaTOOF) for Systems Management Outside of the Telecom Walled Garden

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- 15 Proprietary Implementation of soncli.mcs

PyCS is Open Source – Affero GPL

Software

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soncli.mcs: Moduled Command-Service (Proprietary: Not Open Source)

Integration of Nokia SON SDK with PyCS and bisos.tocsModules.

PyCS Framework

PyCS Facilities: `facter.ics` `sbom.pcs` `gitist.pcs`

WebUI PyCS Player

Blee PyCS Player

PyCS-RO OpenAPI Invoker

PyCS-RO OpenAPI Performer (Django DRF for CSXUs)

PyCS Loadable Modules and Targets (MaTOOF)

PyCS Seeds and Plants

PyCS-RO-RPyC (Invoker)

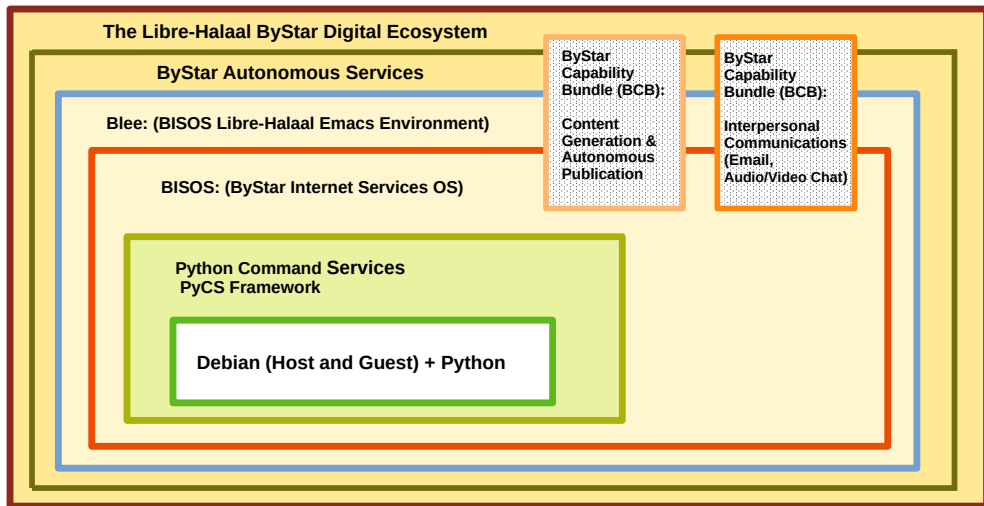
PyCS-Foundation

Py Cmnd Svcs --- PyCS-Core

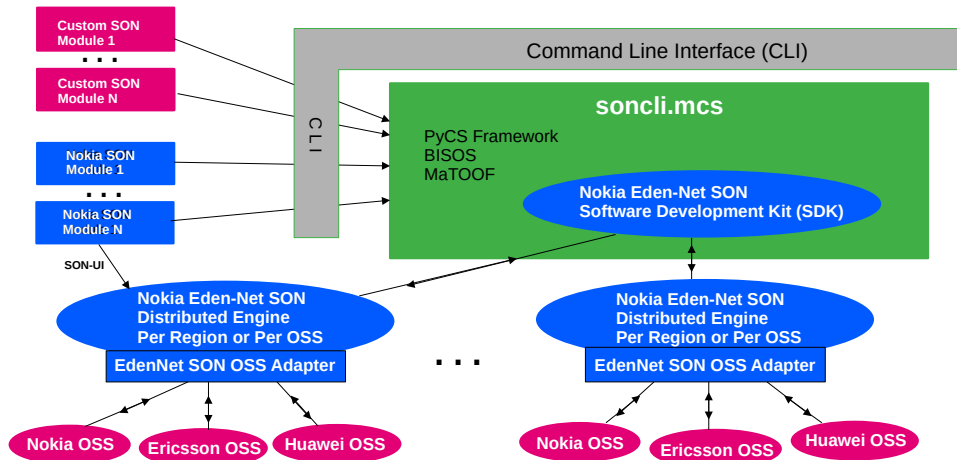
Debian + Python

PyCS as a Substrate of a Digital Ecosystem (ByStar)

The Libre-Halaal ByStar Digital Ecosystem Containment Hierarchy



soncli.mcs -- An After-Market Strategy for Nokia EdenNet SON Development & Deployment



Nokia EdenNet SON Parts List

- SON Modules
- SON Python API
- SON Software Development Kit (SDK)
- SON Web User Interface
- SON Execution Engine
- SON OSS Interface Adapters

soncli.mcs Relevant Parts

- SON Modules
- SON Python API
- SON Software Development Kit (SDK)
(SDK is used as an External Execution Facility — EEF)

Implementation of soncli.mcs is Layered on Top of PyCS

BISOS: ByStar Internet Services Operating System - <https://github.com/bisos>

A Debian based framework for creation of Open Source software-service continuums

PyCS: Python Command Services – <https://github.com/bisos>

A Python framework for creatin of Unified Commands (CLIs) & Services (APIs)

tocsModules: Target Oriented Modules – <https://github.com/bisos-pip/tocsModules>

General facilities for creation of Modules that can be uploaded into CSXUs

CSXUs: Command Services eXecution Units

Commands or Services of PyCS Which Can Be Executed on Command-Line or csPlayers

soncli.mcs: Moduled Command-Service (Proprietary: Not Open Source)

Integration of Nokia SON SDK with PyCS and bisos.tocsModules.

Proprietary Implementation of soncli.mcs

tocsModules: Target Oriented Modules –

<https://github.com/bisos-pip/tocsModules>

General facilities for creation of Modules that can be uploaded into CSXUs

soncli.mcs: Moduled Command-Service (Proprietary: Not Open Source)

Integration of Nokia SON SDK with bisos.tocsModules.

**The Open Source bisos.tocsModules is designed
for Aborbtion of Proprietary SDKs**

Based on bisos.tocsModules Implementation of Proprietary soncli.mcs only involves a very small amount of Python code and effort.

The Bridge: And Then, soncli.mcs Becomes An Integral Part of BISOS.

- 16 The Enablings of soncli.mcs
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- 18 soncli.mcs As An Agentic AI Enabler
- 19 The Poly-SON-Players Model — soncli.mcs is an Additional Modules Player

The Enablings of soncli.mcs

Since SON Modules are Python Modules,

- Modules are loaded through the command line (CLI) and then executed through the SDK (used as an External Execution Facility (EEF))
- Loading and execution of Modules through CLI permits convenient parallel execution at SON Engines
- Modules are surrounded by additional frameworks that enhance usage of modules
- Scope of capabilities of Modules is enhanced

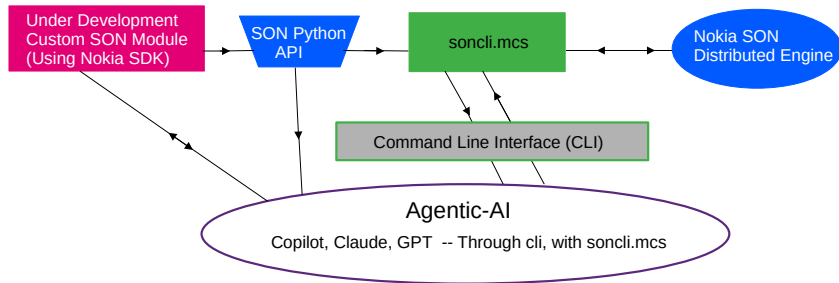
Modules and Target Model of EdentNet SON is then Extended With
The Universal Model of BISOS-MaTOOF (Modules and Target Oriented
Open Fabric) for Systems Management

Immediate and Tangible Benefits of soncli.mcs

- SON Modules Developers Using the SDK Can Load and Run Them Through CLI
 - Subject Under Development Modules To Regression Testing
 - Build CI/CD Pipelines for Custom SON Development
- soncli.mcs is an Agentic-AI Facilitator – Agentic AI Needs CLI
- SON Modules Nationwide (cross OSS) Execution and Reporting
- Broader Systems Management Scope – SON Beyond Cells
- Towards Poly-SON Modules Players

Agentic-AI Needs a CLI

soncli.mcs as an Agentic-AI Facilitator



The Poly-SON-Players Model

soncli.mcs is an Additional Modules Player

- Many SON-Modules have been developed (Think of them as CDs)
- Thus far, there has only been one SON-Modules Player — The EdenNet SON Web-UI (Think of it as a CD-Player)
- Having multiple players is a good thing
- soncli.mcs is a SON-Modules Player (Think of it as another CD-Player)
- As Players, each of SON Web-UI, soncli.mcs and PyCS-Web-UI have their own strength. The EdenNet SON ecosystem becomes richer when there are more Modules-Players.
- soncli.mcs can play any SON-Module, developed by anyone.
- Benefits of soncli.mcs accrue to all Nokia SON Clients.

Using the BISOS Web-UI csPlayer with soncli.mcs

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Abstract Nokia EdenNet SON Parts List – Web-UI-csPlayer

Nokia EdenNet SON Parts List

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Nokia EdenNet SON Parts List

- **SON Web User Interface**
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soncli.mcs Relevant Parts

- SON Modules
- SON Python API
- SON Software Development Kit (SDK)

Web-UI-csPlayer

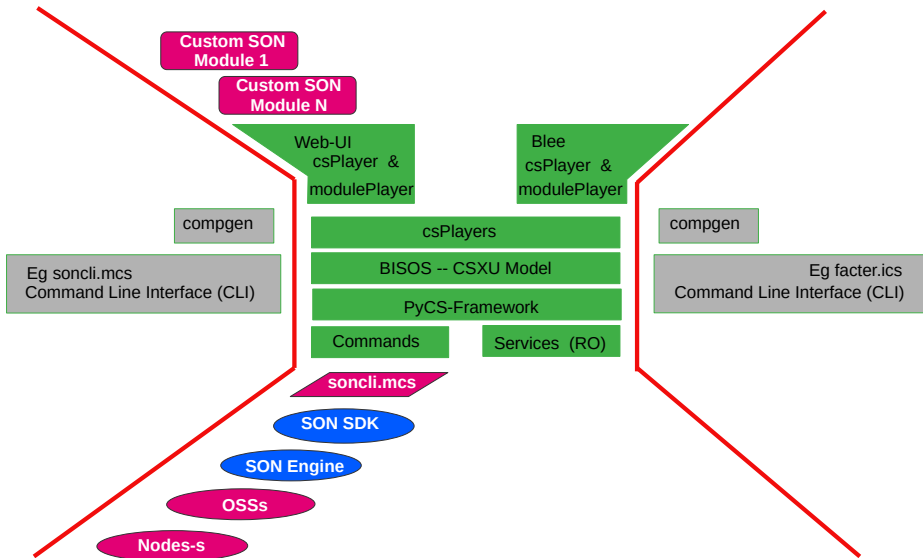
- **Web-UI for “Playing” Modules With soncli.mcs**

The BISOS Web-UI-csPlayer soncli.mcs Functional Spec

- Both: Arrange for Uploading of SON Modules
- Both: Arrange for Execution/Scheduling of SON Modules
- CsPlayer: Arrange for Execution/Scheduling of CSXUs – other than SON Modules Execution
- CsPlayer: Arrange for Parallel Execution of Modules at Multiple Destinations (with airflow)
- CsPlayer: Central Cross OSS Exection of Modules – Nationwide
- CsPlayer: Central (Nationwide) Collection of Execution Results
- Both: Arrange for Textual Selection of Targets
- Nokia: Arrange for Graphical Selection of Targets
- Both: Arrange for Access to Past Results
- CsPlayer: Only Open-Loop, Read-Only Executions (SDK Limitation)
- Nokia: Both Open-Loop and Closed-Loop (Read-Write) Executions
- CsPlayer: Native Open Telemetry (with Grafana Visualization)

Legend:

- Both
Nokia
&
CsPlayer
- Nokia
UI Only
- CsPlayer
UI Only



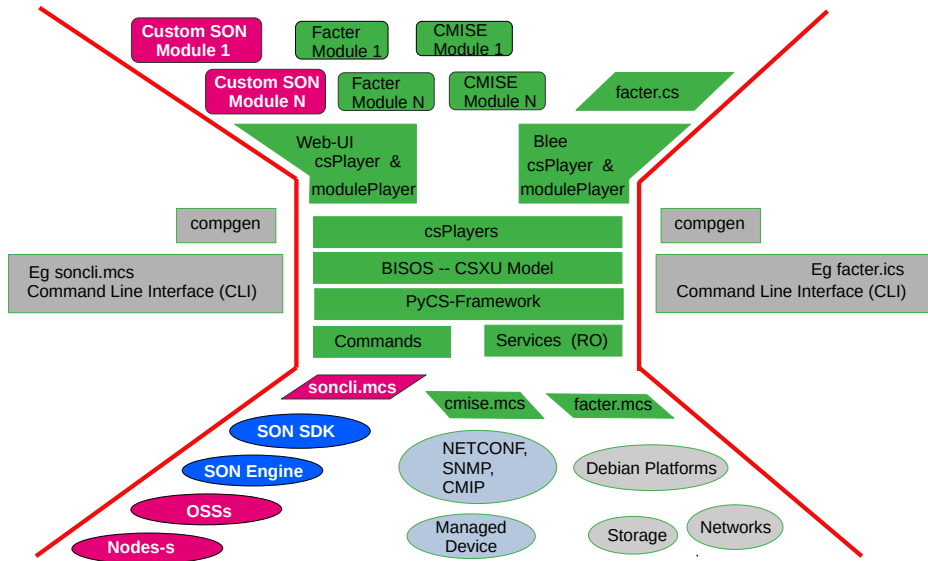
Brining soncli.mcs Into the BISOS-MaTOOF and csPlayer Framework

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- 24 soncli.mcs in the Broader BISOS Systems Management Framework

soncli.cs: A Bridge Into the BISOS/ByStar Environment

- The BISOS-MaTOOF for Systems Management is a very rich framework for managing Networks and Data Centers
- Use of PyCS for implementation of soncli.mcs Allows for SON Modules Developers and Users to profit from the facilities and commonalities of BISOS-MaTOOF for Systems Management.
- This unified architecture that spans Telecom and Datacom is very powerful.

soncli.mcs in the Broader BISOS Sys Management Framework



Our Sales Pitch

25 Requested From You

Our Sales Pitch:

- Adding soncli.mcs to Your Existing Nokia SON EdenNet SON is Big Bang for the Buck
- It is all Open Source (Affero GPL)
- It is all Real
- After Having Adopted soncli.mcs, the Rest of the Benefits Will Become Obvious

Our Requests From You:

- Devote the Time and Energy Needed to Understand What we are Saying.
- Pass It Along. This is All Public. Create Consensus.
- Do Your Due Diligence. Try out the Mentioned Offerings.
- Reachout to Us.